CHAPTER

9

Sterilization and Infection Control

The importance of aseptic techniques using sterilized instruments, disinfectant solutions, etc. coupled with other barriers, has long been emphasized. Unless the instruments and equipment are properly sterilized, a variety of saliva and blood-borne pathogens can easily travel from patient-to-patient. Thus it becomes mandatory for dental professionals to employ effective methods of sterilization and infection control to deliver quality healthcare.

TERMINOLOGY

Sterilization: Sterilization is defined as *"the process by which an article, surface or medium is freed of all living micro-organisms either in vegetative or spore state"*.

Disinfection: Disinfection is the removal or destruction of any micro-organism capable of causing infection.

Three levels of disinfections are differentiated depending on the type and form of micro-organisms destroyed:

i. *High level disinfection:* A process that can kill some, but not necessarily all bacterial spores. It is 'tuberculocidal'; and if the disinfectant is capable of destroying bacterial spores, it is labelled as *'sporicidal'*.

ii. *Intermediate level disinfection:* A process that is capable of killing mycobacterium tuberculosis, HBV and HIV. It may not be capable of killing bacterial spores.

iii. *Low level disinfection:* A process that kills most bacteria, some fungi and some viruses. It does not kill *Mycobacterium tuberculosis* or bacterial spores.

Antisepsis: Antisepsis is a measure used to prevent infection by inhibiting growth of micro-organisms in wounds or tissues. An antiseptic is an agent that inhibits the growth of micro-organisms on living surfaces.

CATEGORIZATION OF INSTRUMENTS

The instruments have been categorized, depending on their contact with body tissues, to determine the preferred mode of sterilization/disinfection.

The categories are as follows:

1. ***Critical items:*** Instruments, which touch sterile areas of the body or enter the vascular system or penetrate the oral mucosa; e.g. curettes, burs, files and scalpels. Dental handpieces are kept in this category. Instruments in this category must be sterilized and stored in appropriate packages. Single use items must be properly discarded.
2. ***Semi-critical items:*** Instruments, which touch mucous membranes but do not penetrate body tissues; e.g. amalgam condensers, saliva ejectors, etc. These items should be sterilized; however, if this is not feasible, high-level disinfection is preferred.
3. ***Non-critical items:*** The items that do not come in contact with the oral mucosa but are touched by saliva or blood contaminated hands; e.g. light switches, countertops and drawer pulls of cabinets. These items should be properly disinfected.

The relation between type of item and its decontamination is tabulated in Table 9.1.

PRESTERILIZATION

The instruments and equipment prior to sterilization includes pre-soaking, cleaning, rinsing, drying and packaging.

a. Pre-soaking, Cleaning, Rinsing and Drying

Pre-soaking of contaminated instruments keeps them wet until adequate cleaning is carried out. This prevents drying of blood, saliva and debris on instruments and facilitates cleaning.

The cleaning process can be either mechanical or manual.

1. *Manual Cleaning*

- Use stiff nylon cleaning brushes
- Use only neutral pH detergents
- Brush delicate instruments carefully and if possible, separate them from general instruments.

Table 9.1: Type of item and methods of decontamination

Disposable items	*Non-critical items*	*Semi-critical items*	*Critical items*
Dispose	Clean ↓ Low level disinfection ↓ Dry and store	Clean ↓ High level disinfection or sterilization ↓ Dry and store	Clean ↓ Sterilization ↓ Dry and store

- Make sure that the instrument surfaces are visibly clean and free from stains and tissues
- Heavy utility gloves should be worn while processing contaminated instruments

2. Mechanical Cleaning

i. *Ultrasonic cleaner:* Ultrasonic cleaning systems (Fig. 9.1) fare better than manual cleaning as they can remove even dried blood and saliva. Moreover, this allows instruments to be cleaned with minimal chance of exposure to body fluids through cuts and punctures and hence, is safer.

Chemicals in the ultrasonic bath solution

- Alkaline phosphatase enzyme
- Buffer (7.0–10.0 pH)
- Glycine
- Deionized water
- Magnesium chloride
- Phosphatase substrate
- Potassium chloride
- Potassium phosphate (monobasic)
- Sodium chloride
- Sodium hydroxide (10N)
- Sodium phosphate, dibasic
- Zinc chloride



Fig. 9.1: Ultrasonic cleaner

An ultrasonic cleaning device should provide fast and thorough cleaning without damage to instruments. The equipment is fitted with a timer to prevent electronic interferences from other electronic systems.

ii. *Washer disinfector:* Physical cleaning with high flow of water (both volume and pressure), spraying over all items, results in effective mechanical cleaning. Temperature is maintained below 45°C to avoid any protein coagulation during the flushing stage.

While disinfection, the temperature of water is elevated gradually to achieve:

- Initial microbial activation phase (temperature 45°– 52°C); endospores germinate and enter negative phase.
- Microbial inactivation phase (temperature 85°–95°C); pathogenic micro-organisms are inactivated or killed. Spores already entered into germinative phase are also killed here (indirectly sporicidal).

Requirements of detergent used in washer disinfector:

- Anti-lipid action at high temperature
- Anti-protein action at low temperature
- Enzymatic action

- Likewise temperature, contact time is also a crucial factor. Lower disinfection temperature requires longer contact times (disinfection phase).

iii. *Enzyme cleaners:* Enzymes are powerful tools for cleaning reusable instruments. A properly selected enzyme cleaner can facilitate pre-cleaning work and improve effectiveness. The key to effectiveness is the selection of the proper enzyme. Enzymes accelerate the chemical reactions of the biological processes and are specific in their catalytic behaviour, such that a protease will work only on proteins. Some enzymes are specific and work on specific proteins while others will work on a broad spectrum of proteins, e.g. Klenzymer cleaners.

The enzymes normally used in instruments cleaning formulations do not attack metals; suited for cleaning many reusable devices.
Many compounds when used in cleaning products enhance effectiveness and stability.
For a surgical cleaning process, protease formulas have been found effective; whereas, for orthopedic surgery, a lipase formula (such as Adi-enzyme) for instrument cleaning is best.
Enzyme cleaner contains proteins in the form of enzymes. If these cling to instrument surface, they can interfere with sterilization process by hiding bacterial contamination. Chemical residues may not be removed thoroughly in the sterilization process.

b. Packaging

The cleaned instruments are dried and packaged in functional sets before sterilization. This packaging protects the instruments from contamination after sterilization and before use.

A variety of packaging materials are available, viz. self-sealing, paper-plastic and peel pouches (commonly used) (Table 9.2).

METHODS OF STERILIZATION

The process of sterilization can be carried out by various methods. The classification of sterilizing agents is tabulated in Table 9.3.

The methods are as follows.

I. Physical Method

A. *Dry Heat*

Dry heat kills by protein denaturation, oxidative damage and the toxic effect of elevated levels of electrolytes.

'The thermal death time' implies the minimum time required to kill a suspension of organisms at a predetermined temperature in a specified environment.

a. ***Flaming/burning:*** The instruments like scissors, tweezers, scalpels, etc. are usually sterilized by this method. The instrument is dipped in alcohol (3 parts of ethyl alcohol and 1 part formalin) and passed over the flame 2–3 times. It is not considered an effective method of sterilization (Fig. 9.2).



Fig. 9.2: Flaming

b. ***Hot air oven:*** Hot air oven can be used to sterilize glassware, forceps, scissors, scalpels and swabs. Cotton, rubber instruments, gloves, etc. cannot be sterilized with this method.

The recommended holding period temperatures are:

- 150°C (two hours) for cutting instruments
- 160°C (45 minutes) for unwrapped and 60 minutes for wrapped instruments
- 170°C (18 minutes)
- 180°C (7.5 minutes)
- 190°C (1.5 minutes)

The oven must be cooled slowly for about two hours before opening the door. Hot air is a bad conductor of heat and its penetrating power is low. The oven must be fitted with a fan for even distribution of air and elimination of air pockets. It should not be overloaded and the materials should be so arranged as to allow free circulation of air in between (Fig. 9.3).



Fig. 9.3: Hot air oven

Table 9.2: Packaging material and sterilization process

Sterilizing process	*Packaging material*
Steam	Paper/plastic pouches, nylon type plastic tubing, sterilization paper, wrapped cassettes, thin clothes
Chemical vapor	Paper/plastic pouches, sterilization paper
Dry heat	Dry heat type nylon plastic tubing, sterilization paper

Table 9.3: Classification of sterilizing agents

Physical	*Chemical*	*Radiation*
A. Dry heat	**A. Chemiclave (chemical vapor: alcohols)**	**1. Non-ionizing radiation**
a. Flaming/burning	a. Ethyl alcohol	a. Infrared
b. Hot air oven	b. Isopropyl alcohol	b. Microwave
c. Glass bead sterilizer	**B. Aldehydes**	
d. Hot salt sterilizer	a. Formaldehyde	
e. Microwave	b. Gultaraldehyde	
	c. Orthophthaldehyde	
	C. Dyes	
	a. Aniline	
	b. Acridine	
B. Moist heat	**D. Halogens**	**2. Ionizing radiation**
a. Temperature below 100°C (pasteurization)	a. Chlorine	a. Ultraviolet
b. Temperature at 100°C (boiling)	b. Chlorine dioxide	b. Gamma rays
c. Steam at atmospheric pressure (steaming)	c. Sodium hypochlorite	c. E-beam radiators
d. Steam under pressure (autoclave)	d. Iodine	**3. Lasers**
C. Filtration (not used in dentistry)	e. Iodophors	a. Carbon dioxide
	E. Phenols	b. Argon
	a. Cresol	c. Nd:YAG
D. Cold	b. Chlorophenol	
a. Desiccation	c. Carbolic acid	
b. Sunlight	**F. Surface active agents**	
	a. Metallic salts	
	b. Surfacine	
	c. Peracetic acid	
	G. Oils	
	H. Gases	
	a. Ethylene oxide	
	b. Formaldehyde	
	c. Betapropiolactone (BPL)	
	I. Miscellaneous	
	a. Plasma sterilization	
	b. Performic acid	
	c. Superdioxide water	

New 'rapid heat-transfer units' are being used:
i. Cox unit (Alpha Scientific)
ii. Guardian 2000 (Dentronix)

c. ***Glass bead sterilizer:*** It is a heat transfer device, wherein glass beads are used as a medium. Glass beads should be less than 1.0 mm in size because larger beads are not effective in transferring heat due to large air spaces between the beads. The instruments to be sterilized are immersed into the heated-up glass beads (temperature 218°C–280°C) and left for a specific period of time (Fig. 9.4).
The time specified for each instrument:

- Absorbent points (butt end first) and cotton pellets: 10 seconds
- Long handled instruments, tips of cotton pliers, blades of scissors: 5 seconds
- Root canal instruments: 5 seconds

d. ***Hot salt sterilizer:*** Hot salt sterilizer uses table salt in place of glass beads as heat transferring agent. It is used to sterilize absorbent points, broaches, files, reamers and other root canal instruments.
The instruments desired to be sterilized are put into the sterilizer and left for a period of time, specific for that instrument. Immerse instrument at least a



Fig. 9.4: Glass bead sterilizer

quarter inch below salt's surface. It consists of a metal cup in which table salt is kept at a temperature of 425°F– 475°F (218°C–246°C).

Both glass bead sterilizer and hot salt sterilizer (mainly used for endodontic instruments) are considered to be an auxiliary method of sterilization, as they have been proved inefficient in killing spores.

Advantages

- Use of table salt (readily available)
- Table salt does not cling to the wet instruments, as glass beads and molten metal do.
- Cost effective

e. ***Microwave sterilization:*** Microwave energy has been utilized in the past as a means of sterilizing food products, etc. Microwave sterilization of dental instruments could not become popular because the metals are heated only slightly, unless they are magnetic. An auxiliary heat source is required with the microwaves in order to produce heat, which in turn will sterilize the instruments. The metallic instruments, under microwave, develop a corona discharge or arc, which can melt the instrument tips, if placed together.

The microwaves vaporize the liquid sterilant and the instruments are exposed to vaporized sterilant and/or microwaves. The instruments are placed in a pressurized atmosphere. Alternatively instruments can directly be exposed to microwaves in microwave oven. The drawbacks of this procedure are (i) the instruments need to be rotated three-dimensionally, (ii) microwave absorbers are required within the oven to prevent arcing.

Advantages of dry heat

- Maintenance of sharp edges of the cutting instruments. Practically, dry heat is the best method for sterilization of burs.
- No corrosion of instruments.
- Less expensive than steam autoclaves.
- Industrial forced draft ovens usually provide a larger capacity at a reasonable price.
- Rapid cycles possible at high temperature.
- Packs are dry after cycle.

Disadvantages of dry heat

- Prolonged sterilization time is required because of poor heat conduction by air and poor penetration properties of dry heat.
- High temperature may damage more heat sensitive items, such as rubber or plastic goods.
- Sterilization cycles are prolonged at lower temperatures.
- Heavy loads of instruments, crowding of packs and heavy wrapping easily defeat sterilization.

Caution

- Instruments should be pre-dried
- Instruments should not be added during the sterilization cycle.

B. *Moist Heat*

Moist heat (steam, boiling water) kills by denaturation and coagulation of protein. The advantage of steam is that the latent heat condenses on a cold surface, raising the temperature of that surface. In a complete moisture-free atmosphere, bacteria are resistant to heat (because oxidation of cell contents requires higher temperatures than that needed for coagulation of proteins).

The procedures employed are:

a. ***Temperature below 100°C:*** It is used for pasteurization of milk. Either 63°C for 30 minutes (holder method) or 72°C for 15–20 seconds (flash process) followed by cooling rapidly to 13°C or lower. All non-sporing pathogens are destroyed. Vaccines or non-sporing bacteria are inactivated at 60°C for one hour in vaccine baths. *Staphylococcus aureus* and *Streptococcus faecalis* require 60 minutes at 60°C; 80°C for 5–10 minutes destroy the vegetative forms of all bacteria and yeasts. Poliomyelitis virus requires 60°C for 30 minutes and hepatitis virus requires 60°C for 10 hours.

b. ***Temperatures at 100°C:*** Vegetative bacteria are killed almost immediately at 90–100°C, but not spores (Fig. 9.5). Hard water for boiling should not be used. Sterilization can be promoted by the addition of 2.0% sodium bicarbonate to water.

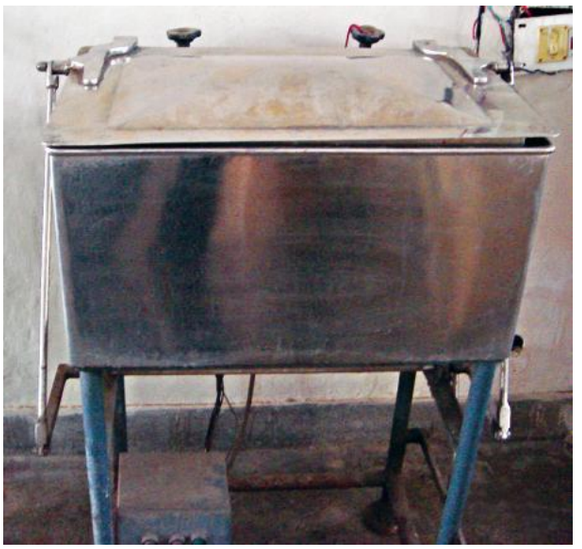


Fig. 9.5: Sterilization by boiling

Material should be immersed in water and boiled for 10–30 minutes without opening the lid of sterilizer. Time periods up to 24 hours may be necessary to kill bacterial spores and even this prolonged time will not kill many viruses. For this reason, boiling water is not recommended for sterilization of tissue penetrating instruments.

c. ***Steam at atmospheric pressure:*** In case of steamer, the instruments are in touch with steam only. It consists of a vertical metallic container having a perforated tray in the middle. The water is filled under it, which boils and produces steam. The steam at 100°C sterilizes the instrument (an exposure of 90 minutes usually ensures sterilization).

d. ***Steam under pressure (autoclave):*** The principle of autoclave is that, water boils when its vapor pressure equals that of the surrounding atmosphere. Saturated steam has greater penetrating power. When steam has greater penetrating power, it condenses to water and gives up its latent heat to that surface. The condensed water ensures moist conditions for killing the microbes.

Various types of autoclaves are shown in Fig. 9.6A–C.

The recommendations for temperature and holding period are:

- 121°C—15 minutes at 15 lb pressure
- 136°C—3 minutes at 30 lb pressure for unwrapped and 5–12 minutes for wrapped instruments.

Instruments should be removed only after drying completely as wet packages are considered non-sterilized.

Articles that can be sterilized in the autoclave include most culture media, saline and other articles not damaged by high temperatures, viz. syringes, needles, dressings, sponges, gowns, rubber gloves, tubing, aprons, etc.

Autoclaving is damaging for almost all high speed dental handpieces. Without prior treatment of oil emulsion, burs are also destroyed in autoclaving.

Flash sterilization (rapid cycle autoclave): Flash sterilization is a modification of conventional high vacuum steam sterilization, used for carrying out rapid chairside sterilization of critical items. Instruments are placed unwrapped at 132°C for 3–4 minutes at 27–28 lb pressure. The sterilization time may vary with the type of sterilizer and the type of item (i.e. porous vs non-porous items).

The method is acceptable for processing pre-cleaned instruments that cannot be packaged, sterilized and stored before use.

It is used when there is insufficient time to sterilize an item by the preferred method.

Flash sterilization is not considered as effective as the routine packaged sterilization and thus should not be used for convenience.

Precautions while operating the steam sterilizers

- Turn off the value
- Steam should be allowed to dissipate from the sterilizer; until then do not open the door.
- The sterilized goods (pack) should not be placed on a cold surface, because the heat will condense upon contact with the surface and may contaminate the sterilized pack.

Advantages of autoclaves

- Rapid and effective method for sterilizing cloth packs and towel packs. Other methods are not suitable for processing cloth packs.
- Good penetration of heat.

Disadvantages of autoclaves

- Heat sensitive items like plastic and rubber cannot be autoclaved.
- Tend to rust carbon steel instruments and burs.
- Steam appears to corrode the steel neck and shank portions of some diamond instruments and carbide burs. To avoid or minimize the corrosive action of steam on metals, ammonia, dicylohexyl ammonium nitrite or cyclohexylamine and dicyclomine or 2.0% sodium nitrite can be used.

C. Filtration

The method is not used in dentistry, though it is used for sterilizing fluids, which cannot be heated.



Fig. 9.6: Types of autoclaves

D. *Cold*

The cold method is used to retard microbial growths in foods and to preserve biological materials. It acts by:

- Increasing viscosity of the proteins
- Impeding diffusion of toxic products from the cell
- Increasing enzymatic activity
- Formation of ice crystals within and outside the bacterial cell, which mechanically ruptures the cell. Temperatures around 0°C have more inhibitory effect than sub-zero temperature.

a. *Desiccation:* This method is unreliable and is only of theoretical interest.

b. *Sunlight:* The microbicidal activity of sunlight is due to the presence of ultraviolet rays. Sunlight is not sporicidal; hence, it does not sterilize.

II. Chemical Method

Various chemical agents are used to disinfect, mainly the operating surfaces. A few agents can achieve sterilization also.

The desirable properties of disinfectant are:

- Non-corrosive
- Readily available
- Pleasant flavour
- Non-irritating
- Effective on most micro-organisms
- Economical.

The chemical agents used for disinfection act in various ways:

- Protein coagulation
- Disruption of cell membrane
- Removal of free sulfhydryl groups, which are essential for functioning of the enzymes
- Substrate competition

The levels of activity of various chemicals for disinfection are tabulated in Table 9.4.

Office disinfection involves using a liquid chemical at room temperature on submerged instruments or operatory surfaces. If the chemical is not sporicidal (iodophor, phenol, alcohol/phenolics, etc.), it is called disinfectant.

If the chemical is sporicidal, it is called a sterilant (2.0% or 3.0% concentration of glutaraldehyde) and may be expected to achieve sterilization; however, only after 10 hours contact time.

Various factors, which affect germicidal activity of chemical agents are:

- Effective concentration of agent
- Time of exposure, kind and number of organisms, type of material to be disinfected, presence of organic matter.
- Increased temperature results in increased lethal activity.
- Some chemical agents have more activity in acidic pH (anionic detergents) and some in basic pH (cationic detergents).
- Some agents may combine with organic matter and become ineffective.

A. *Chemiclave (Chemical Vapor: Alcohols)*

Chemical vapors are used to sterilize dental instruments, especially instruments which are prone to rust. The active chemical ingredients include

Table 9.4: Chemicals for disinfection

Level	*Use*	*Examples*
Low level	Surfaces without blood	Phenol and iodoform
Intermediate level	Surfaces with blood	Alcohol, Chlorines, Phenol, Iodoform
High level	Immersion	Glutaraldehyde, o-phthalaldehyde

72.38% ethanol and 0.23% formaldehyde (vaposteril). The water content is below 15% level, above which rust, corrosion and dullness of metal occurs.

The chemical vapor sterilizers are preferred for instruments, which cannot be heated beyond 132°C and instruments prone to rust.

The sterilization cycle of the chemiclave is at 132°C at 20 lb pressure for 25–30 minutes. Instruments can be sterilized in both wrapped and unwrapped state in 20 minutes. The main advantage of chemical sterilizers is that instruments do not corrode nor they get blunted.

However, penetration of the chemical vapors is not as effective as that of steam in the autoclaves.

- ***Ethyl and isopropyl alcohol:*** Ethyl alcohol and isopropyl alcohol are the most frequently used chemical disinfectants; used mainly as skin antiseptic and act by denaturing proteins (Fig. 9.7). There is no action on spores or viruses. It is generally used at a concentration of 60–70% in water. Ethyl alcohol, on standing, oxidizes to acetic acid and acetaldehyde, which are corrosive to metals. Isopropyl alcohol is a better fat solvent, more bactericidal and less volatile than ethyl alcohol. Methyl alcohol is effective against fungal spores and is used for treating cabinets and incubators. Alcohol is poorly effective against all micro-organisms and not at all effective against spore formers and is not an accepted method for disinfection of surfaces and instruments.



Fig. 9.7: Ethyl alcohol and isopropyl alcohol

Advantages

- Fast turn-around time.
- Protection of carbon steel instruments from rust or corrosion.
- Availability of dry, immediately usable instruments.

Disadvantages

- Items sensitive to elevated temperature will be damaged.
- Bad odour is released when chemicals are heated.
- Limited availability of apparatus and high cost.

Caution

- Load only dry instruments.
- Check the door for any leakage.

B. *Aldehydes*

a. *Formaldehyde:* Formaldehyde in aqueous solutions is markedly bactericidal, sporicidal and virucidal (Fig. 9.8A). Formaldehyde gas is used for sterilizing instruments, catheters, fumigating wards, and laboratories. Clothing, bedding, furniture and books can also be disinfected. The irritant vapor can be nullified by exposure to ammonia vapor after disinfection is completed. It mainly acts on protein molecules of the micro-organisms.

b. *Glutaraldehyde:* Glutaraldehyde is similar to formaldehyde, but less toxic and less irritant (Fig. 9.8B). It is used to disinfect rubber anesthetic tubes, face masks, plastic endotracheal tubes, metal instruments and polythene tubing. Glutaraldehyde kills bacterial spores, fungi and almost all viruses. Prior to use, the instruments must be thoroughly rinsed with sterile water. Disinfection occurs in 10–30 minutes. Various preparations of glutaraldehyde are available:

- 2.0% acidic (60°C for one hour)
- 2.0% alkaline (room temperature for 10 hours)
- 2.0% alkaline (room temperature for 6.75 hours with phenolic buffer)
- 2.0% neutral (room temperature for 10 hours)

Glutaraldehyde is not recommended for sterilization because of the instability of activated solutions, problems of dilution and inability to monitor sterilization.

Disadvantages

- Corrosion of easily oxidized metals
- Need for fresh solution for each sterilization/disinfection process.

c. *Orthophthalaldehyde:* The solution is available in 0.55% active orthophthalaldehyde concentration; shows excellent microbicidal activity.

Advantages

- Shorter processing time
- Non-irritant to eyes and nasal passages
- Excellent stability over a wide range of pH (3–9)
- Weak; barely perceptible odor.



Fig. 9.8: **(A)** Formaldehyde, **(B)** Glutaraldehyde

Disadvantages
• Tends to stain proteins gray; thus needed to be handled with caution, avoiding exposure to skin. • High cost.

C. Dyes

The aniline and acridine dyes are used as skin and wound antiseptics. Both are bacteriostatic and are effective against gram-positive organisms. If impregnated in gauze, they are slowly released in a moist environment.

Disadvantages
• May discolor surfaces. • Must be prepared daily.

D. Halogens

The commonly used halogens are:

a. ***Chlorine:*** Chlorine in aqueous form is used as disinfectant of water supplies, swimming baths, food and dairy industries. It is bactericidal and has a wide spectrum of activity against viruses. Organic chloramines are used as antiseptic for dressing wounds (Fig. 9.9A).

b. ***Chlorine dioxide:*** Chlorine dioxide has the ability to disinfect instruments in one to three minutes. The solution requires no rinsing and leaves no residue after use. Sterilization may take six hours at room temperature. The solution/compound has been reported to be non-toxic, non-irritating and non-sensitizing (Fig. 9.9B).

c. ***Sodium hypochlorite:*** Sodium hypochlorite is more suitable for surface disinfection than for instrument sterilization, because of its highly corrosive action on metals. (It is virucidal, bactericidal and tuberculocidal.) Dilution of 1:5 to 1:1 are generally recommended. Higher concentration can achieve disinfection within three minutes. It is the least expensive of the surface disinfectants (Fig. 9.9C).

Disadvantages
• Corrosive factor. • Solution tends to be unstable and should be prepared daily. • Strong, unpleasant odor. • Plastic chair covers have a tendency to crack after prolonged use.

d. ***Iodine:*** Iodine in aqueous and alcoholic solutions has been used as skin disinfectant. It is an active bactericidal and moderately active against spores and viruses.

e. ***Iodophors (iodine mixed with a detergent):*** Iodophor is a broad-spectrum disinfectant. One of the inherent advantages of iodophores is slow release of the elemental iodine, enhancing the bactericidal activity. A surfactant carrier keeps a surface moist to prolong the release.

Iodophors have an in-built color indicator. When the solution is fresh, an amber color is present. With age, the solution changes to light yellow, indicating the loss of iodophor molecules. The iodophor compound is to be used solely as a disinfectant. Its sporicidal capabilities have not been established.

Disadvantages
• Vapor toxicity • Irritation to skin and eyes • Discoloration of some metals • Expensive



Fig. 9.9: Halogens as disinfectant

E. *Phenols*

Phenols are effective disinfectants causing cell membrane damage and subsequent lysis. Low concentrations precipitate proteins, inactivating oxidases and dehydrogenases. None of the phenolic compounds are sporicidal at room temperature, but they are active against many viruses, fungi and bacteria.

Phenolic compounds are used for disinfection of walls, floors or other surfaces that do not come into contact with body tissues.

Chlorhexidine (Hibitane) is a relatively non-toxic skin antiseptic, active against gram-positive organisms and fairly effective against gram-negative ones. Aqueous solutions are used in the treatment of wounds.

Disadvantages

- Cause damage to some plastics
- Corrode certain metals, such as brass, aluminium and carbide steel (Corrosive effect is neutralized by mixing them with sodium bicarbonate.)

F. *Surface Active Agents*

Surface active agents are used as wetting agents, detergents and emulsifiers. They are classified into anionic, cationic, non-ionic and amphoteric groups. The cationic surface active agents are important antibacterial agents. They act on phosphate groups of cell membrane, causing the membrane to lose its semipermeability; subsequently, cell proteins are denatured.

Cationic agents in the form of quaternary ammonium compounds are bactericidal, active against gram-positive organisms and to a lesser degree gram-negative organisms. They also act on spores and most viruses. Examples are: Cetyl trimethyl ammonium bromide (cetavlon or cetrimide) and benzalkonium chloride. Anionic agents (soap) and organic matter reduces their efficacy.

Anionic compounds (soap) have moderate activity. Soaps from saturated fatty acids are more active than those from unsaturated fatty acids.

a. *Metallic salts:* Salts of mercury, silver, copper are used as disinfectants. They are protein coagulants. Mercuric chloride is highly toxic as compared to organic compounds (thiomersal, phenyl mercury nitrate and mercurochrome), which are used as mild antiseptics. Copper salts are used as fungicides.

b. *Surfacine:* Surfacine has persistent activity (up to 13 days) that may be used on animate or inanimate surfaces. It incorporates a water-soluble antimicrobial compound (silver iodide) in a surface-immobilized coating (polyhexamethylene biguanide). Micro-organisms accumulate silver until the toxicity level is exceeded; dead organisms eventually lyse and detach from the surface. It can be applied on the surface by brushing, dipping or spraying without prior surface treatments. Their antimicrobial efficacy has not been thoroughly documented.

c. *Peracetic acid:* Peracetic acid in aqueous solution is an effective germicide against the wide spectrum of micro-organisms; however, it is corrosive. At higher concentration, it is sporicidal also.

G. *Oils*

Hot oil baths have been used to sterilize metallic instruments. The oil can reach a temperature of 175°C; 15 minutes of submersion is necessary for sterilization. Longer period of submersion has been more effective (up to one hour).

H. *Gases*

a. *Ethylene oxide:* Ethylene oxide has a boiling point of 10.7°C. This is considered a highly penetrating, inflammable and explosive gas. It is rendered non-inflammable for hospital use when mixed with carbon dioxide or fluorinated hydrocarbon, such as freon. The objects are placed in a chamber, exposed to the gas for a certain period of time (usual exposure time ranges from 2 to 12 hours). Humidity and temperature are critical factors in the destruction of micro-organisms during gas sterilization. The chamber is generally maintained at 40 to 80% humidity, temperature ranging from 50°C to 60°C and gas concentration between 200 and 800 mg/dL.

The features of importance are:

- Ability to alkylate the carboxyl, hydroxyl and sulfhydryl groups in the protein molecule; also reacts with DNA and RNA.

Caution

The oil can defeat sterilization, so handpieces should be cleaned but not oiled before ethylene oxide sterilization. The cost of the equipment and long sterilization time make this method impractical.

Advantages

- Preferable for materials sensitive to heat.
- Complete penetration.

Disadvantages

- Time of sterilization and ventilation is long.
- Toxic (carcinogenic, flammable and explosive).
- Need an aeration period after the process due to formation of ethylene chlorohydrin.

- Toxic effect (autogenicity and carcinogenicity); all items sterilized by this method must be aerated following exposure.
- Used for sterilizing heart-lung machines, respirators, sutures, dental equipment, books, clothing, glass, metal and paper surfaces, plastics and food-stuffs, etc.

b. *Formaldehyde:* It is used for fumigation of operation theatres (150 gm of potassium permanganate to 280 mL formalin for an average room). Heat resistant vessels should be used. The doors should be sealed and left unopened for 48 hours. It is less popular than ethylene oxide because of its odor.

Advantages
• Preferable for materials sensitive to high heat.
• No need for ventilation of materials after sterilization.

Disadvantages
• Toxic and carcinogenic.
• Cannot be used for the sterilization of liquids.

c. *Betapropiolactone (BPL):* Betapropiolactone, having boiling point of 163°C, is a gas of low penetrating power with rapid biocidal action but also has carcinogenic activity. In the concentration of 0.2%, it is used for sterilization of biological products. The BPL is capable of killing all micro-organisms and is very active against viruses.

I. *Miscellaneous*

a. *Plasma sterilization:* Plasma is basically an ionized gas. Plasma sterilization involves UV irradiation, photodesorption and chemical etching. The spores are made up of atoms like carbon, oxygen, hydrogen, nitrogen, etc. The radicals react with these atoms to form compounds like carbon dioxide, which can be subsequently flushed out. When the organism loses atoms intrinsic to its survival, it dies.

Advantages
• The process is carried out usually at room temperature and hence poses no dangers associated with high temperature (unlike autoclaves).
• It does not involve any chemical and hence non-toxic (unlike ethylene oxide)
• Time of treatment is fast and of the order of one minute or less.
• It is versatile and can sterilize almost any material and any shape.

Disadvantage
• Not a proper method for the sterilization of liquids

b. *Performic acid:* Performic acid is produced by using two solutions—hydrogen peroxide and formic acid. It is fast acting against spore forming bacteria and is being used in an endoscope-reprocessing system marketed as 'Endoclens'.

c. *Superdioxide water:* Superdioxide water contains hypochlorous acid at a concentration of approximately 144 mg/L and free chlorine radicals. It is non-toxic to biological tissues and non-corrosive to metallic instruments with a broad-spectrum activity.

III. Radiation Method

A. *Non-Ionizing Radiations*

The non-ionizing radiations consist of rays with wavelength longer than those of visible light. These rays are absorbed as heat. Sterilization achieved by non-ionizing radiations are:

a. *Infrared radiation:* The heat generated by these radiations and subsequently oxidation leads to killing of micro-organisms. Five to ten minutes exposure is sufficient to sterilize all the equipment. The infrared rays can also be used for monitoring the sterilized instruments.

b. *Microwave sterilization:* Explained on page 167.

B. *Ionizing Radiations*

The ionizing radiations can be either particulate or electromagnetic. The particulate radiations are cathode rays having high energy electrons. These are usually used in treatment of cancers.

The electromagnetic radiations are of short wavelength (UV-rays, γ-rays and cosmic rays). Ionizing radiations are highly lethal to DNA and other vital cell constituents. There is no appreciable increase in temperature and the process may be referred as 'cold sterilization'.

The types of ionizing radiations used for sterilization process are:

a. *Ultraviolet radiations:* The bactericidal range of ultraviolet wavelength is 240–280 nm. The rays are absorbed by protein and nucleic acid of the organisms. A chemical reaction is set within the cell, which leads to its death. These rays are used to clean the surface environment. The air, water and the surroundings are sterilized by this method. Ultraviolet radiation is used to disinfect enclosed areas, such as entryways, hospital wards, operation rooms, laboratories for storage and maintenance of previously sterilized materials. UV rays have much lower energy and poor penetrability as compared to X-rays (Fig. 9.10).

b. *Gamma rays:* Gamma ray is a commonly employed method for industrial sterilization of heat sensitive



Fig. 9.10: Ultraviolet unit

products. The source of gamma rays (Co-60) is housed within a reinforced concrete chamber with 2.0 meter thick walls. Articles being sterilized are passed through the irradiation chamber. Gamma radiation easily reaches all parts of the object due to its high penetration ability. The items can be pre-packed in hermetically sealed packages, impermeable to micro-organisms, before sterilization. The sterile shelf life of these supplies is up to the point of use. These radiations are measured as 'rads'. Usually, 2.5 megarads are required for effective sterilization.

c. ***E-beam irradiation:*** E-beam irradiation method, used for the sterilization of medical devices, has many advantages, like being safe, having no emission and high speed processing. It is similar to gamma irradiation as being an ionizing energy; but, the difference is its high dosage rates and low penetration. Another difference is that the e-beams have a source of electricity producing high charge of electrons. E-beam causes a change in the chemical and molecular bonds, subsequently destruction of DNA chain of the reproducing bacteria. The process is cost-effective but the construction of the e-beam sterilization institution is expensive.

C. Lasers

Carbon dioxide laser has been widely used as one mode of sterilization. It is considered effective in complete eradication of the bacteria from the instrument surfaces. Argon laser and Nd:YAG have been used to sterilize operative instruments. Argon Laser is considered better than the other two lasers.

Tabler 9.5: Recommended sterilization procedures for commonly used articles

Instruments	*I Choice*	*II Choice*	*III Choice*
Metal instruments	Moist heat	Dry heat	Chemical
Surgical burs	Moist heat	Dry heat	Chemical
Surgical handpieces	Moist heat	Chemical	—
Air rotor handpieces	Moist heat	Chemical	—
Air rotor cord	Gas	—	—
Micromotor cord	Gas	—	—
Rubber tube suction	Moist heat	Chemical	Gas
Cautery tip	Gas	Chemical	—
Mouth mirror	Moist heat	Chemical	Dry heat
3-way syringe	Moist heat	Chemical	Dry heat
Glass slab	Gas	Chemical	Chemical
Sutures	Gas	Chemical	—
Needles	Moist heat	Dry heat	Chemical
Gauze, gloves	Moist heat	—	—
Suction tip	Moist heat	Dry heat	Chemical
Light cure tip	Moist heat	Chemical	Dry heat
Drapes, gowns	Moist heat	—	—

CAUSES OF STERILIZATION FAILURE

Failure of sterilization is a major concern of the operator. The failure can be because of varied reasons; the main are as follows.

i. ***Improper cleaning:*** The instruments should be thoroughly cleaned before sterilizing. Dental materials, blood or even saliva may not allow the sterilizing agent to reach at the site. Preferably the cleaning should be carried out with detergent; alternatively wash in running water.

ii. ***Improper placement:*** The instruments to be sterilized should not be packed tightly, or one above the other. Appropriate space should be there between the instruments so as to allow the sterilizing agent reach all the surfaces.

iii. ***Improper timings:*** The sterilizing agent should contact the surface of the instrument for a given period of time depending upon the unit, which is used and the instruments/objects, which are sterilized. The sterilizing time includes safety factor time, which ensures the total killing of micro-organisms.

iv. ***Malfunctioning of the unit:*** Sometimes the unit, which is used for sterilization may have manufacturing defect, which leads to failure. For

example, during autoclaving, the pressure and/or the temperature displayed may not be correct and the operator believes the false readings.

CONTROL OF STERILIZATION

The operator must check the appropriateness of sterilization, i.e. whether the required temperature is achieved or the object is free of micro-organisms.

Various tests and techniques have been designed to ensure authenticity of the sterilization process. The tests employed are:

i. ***Autoclave tape:*** The tape is impregnated with a dye; the color white is changed to brown/black depending upon the temperature achieved. This test is only an indication of the temperature and cannot be taken as surety of sterilization. Similar tapes are also designed, which indicate the penetration of steam under pressure.

ii. ***Thermocouple:*** The thermocouple probes are designed specifically to monitor the temperature and the time during sterilization. The probes can be inserted into the article to be sterilized and the meter indicates the temperature and the time.

iii. ***Microbial tests:*** In this test, a particular species of bacteria (usually *Bacillus stearothermophilus* for steam and chemical vapour, *Bacillus subtilis* for dry heat and ethylene oxide) is used as a test organism. Paper strips impregnated with around one million bacterial spores are dried and put along the object to be sterilized. After sterilization process is over, the strips are incubated for 5–7 days. Though this method is perfect to test sterility; however, practically not suitable because of the time lapse for knowing the exact status of the sterilization.

iv. ***Chemical indicators:*** Certain chemical agents, at certain temperature and particular time, lead to color change, which shows the sterilization achieved. The chemical shows red for danger; orange for caution and green for complete sterilization. Different types can be used for steam sterilizers and for autoclaving.

STERILIZATION OF HANDPIECES

The dental handpiece and turbine due to constant contact with oral fluids are contaminated on both external and internal surfaces. Micro-organisms are capable of forming biofilms on the internal surface of handpieces and waterlines. Contamination of the internal parts of the handpiece may originate from saliva aspirated during treatment or from the air and water supply of dental units, where micro-organisms may build up in biofilms.

A range of microbiological flora has been identified in dental unit waterline samples.

Rinsing out waterlines of the unit with Tween 80 and Ponceau 4R dye gives a marked reduction in micro-organisms. Hydrogen peroxide in a concentration of 3.0% is also helpful in reducing the flora.

The sterilization of handpieces can be achieved by the following methods:

a. Steam Sterilization

Steam sterilization of handpieces is one of the routinely used methods. The cleaning and lubricating is performed as prescribed by the manufacturer.

Cleaning the fiberoptics with detergent solution or any other suitable organic solvents may prove useful. For handpieces with a metal-bearing turbine, the handpieces and the sheath is cleaned with running water and detergent. See manufacturer's directions for further cleaning and lubrication before and after sterilization. Cover the handpiece in a sheath and autoclave.

The steps followed are:

- Leave the handpiece attached to the hose after operation.
- Wipe away visible debris from the handpiece.
- Flush water and air lines with the air/water system.
- Remove the handpiece from the hose and clean the outside thoroughly.
- Clean/lubricate internal portions as directed by the manufacturer. Reattach to the air/water system and blow out excess cleaner/lubricant with a vacuum line. (Depending on the handpiece, it is to be lubricated before and after sterilization.)
- Wipe away excess lubricant from the outside. If using fiberoptic handpieces, clean the lubricant from the fiberoptic connecting interface as directed by the manufacturer.
- Follow the manufacturer's instructions for the type of heat sterilizer (e.g. steam, autoclave, chemical vapor) and maximum temperature that can be used.
- Package the handpiece in the proper bag for the type of sterilizer being used and heat process following the sterilizer instructions.

b. Chemical Vapors

Chemical vapors work well with ceramic-bearing handpiece; however, chemicals may impair functioning of other hand pieces.

c. Ethylene Oxide Gas

Ethylene oxide gas is commonly used for handpieces. Internal and external cleaning is important so as to facilitate ethylene oxide gas to penetrate the handpieces. However, oil left in the handpieces can impair sterilization.

The ethylene oxide exposure for at least 12 hours is required for an effective sterilization of handpieces. A device, which combines flushing, lubrication and autoclaving has been marketed as 'Sterimax'.

STERILIZATION OF BURS

Dental burs are heavily contaminated after use. Presently, single-use burs are recommended to overcome cross-contamination.

Several options exist in handling contaminated dental burs prior to sterilization.

- Bur is covered with wet gauze using a disinfectant.
- Place in an ultrasonic cleaner or washer disinfector using separate bur holder.
- Hand scrub with a mild disinfectant.

For autoclave sterilization, burs can be protected from rusting by keeping them submerged 2.0% sodium nitrite solution.

- Add 20 mg of sodium nitrite crystals in 1 L of distilled water to prepare 2.0% sodium nitrite (not nitrate) solution.
- After ultrasonic cleaning, rinse the burs and place them in a small glass or metal beaker with a perforated lid.
- Fill the beaker with fresh nitrite solution ensuring complete submersion of burs.
- Autoclave with a normal sterilization cycle.
- Fluid from the container is discarded through the perforated lid and burs are transferred to a sterilized bur holder or tray using a sterile forceps.
- Before use, if needed, rinse off burs with sterile water to wipe away any residue left on burs.

INFECTION CONTROL

The 'standard precautions' as standard of care for minimizing the risks of cross-infection between patients and the staff have been widely accepted. The standard precautions emphasize the need for prevention of exposure to and transmission of not only blood-borne pathogens, and also other pathogens encountered in oral health care settings. The essence of precautions is the placement of impermeable barriers between health care personnel and their patients' potentially infectious body fluids; and also use of methods to reduce injury and accidents involving sharp, body fluid contaminated instruments.

Use of gloves, masks, protective eyewear, overgarments, plastic barriers, etc. provide a healthy environment.

ENVIRONMENT OF DENTAL OPERATORY

The dental operatory may get exposed to various types of microbial contaminations. The features, which contaminate the environment are:

a. ***Air-borne contamination:*** Air-borne contaminants, viz. aerosols, spatter, etc. are commonly encountered in dental operatory.
b. ***Digital contamination:*** The methods to minimize direct contact with various surfaces are required to avoid digital contamination (walls, partition, cabinets, etc. are all contaminated).
c. ***Cross-infection:*** In the dental settings, evidence of oral or systemic cross-infections is difficult to obtain, because such patients may have contacted infections elsewhere, before or after having a dental treatment. Infected patients usually are not aware of the source of their infection. A thorough history and previous medical records are helpful.
d. ***Patient vulnerability:*** Infection risks for dental patients have not been investigated as properly as for hospital patients; appear to be less vulnerable. If infection control practices are used, transmission of infection from dental patients is rare.
e. ***Personnel vulnerability:*** When dental personnel experience exposure to saliva, blood and possible injury from sharp instruments, they are vulnerable to infections. Use of proper protective barriers can minimize the contaminants.

The operating area/chamber is to be sterilized for better clinical outcome. The complete eradication of bacteria may not be possible in operating room. The source of bacteria can be from air, patients attending the area, articles in the room, working equipment, etc.

The methods used to keep the operating area bacteriologically safe are:

i. Air flow is filtered; circulating purified air and continuously removing the contaminated air. The patients are allowed in the area 'off shoes' and preferably in sterilized gowns.
ii. Standard cleaning and disinfection procedures should be followed, which minimizes the bacterial growth.
iii. Fumigation of the area with formalin fumes. Since formalin is pungent, the procedure is carried out overnight in a closed environment. The room is kept closed for 12–14 hours and is ready for use once all the fumes are out. Formalin is usually neutralized with ammonia (2 hours contact) prior to use of the room. Formalin use (fumigation) is obsolete in many countries because of its potentially carcinogenic nature. Frequent use of inhalation is hazardous. Newer agents are being used to sterilize the room.

iv. Newer agents for sterilization of operating area are:
- *Bacillocid rasant* (composition: Glutaraldehyde, benzalkonium chloride and dimethyl ammonia). It is formalin-free disinfectant with excellent cleaning properties and is cost-effective. It provides complete asepsis within one hour; closing the room for 12 hours is not required.
- *Virkon:* It is non-aldehyde compound (composition potassium monosulfate, sulfamic acid and sodium dodecyl-sulfonate), considered safe virucidal and fungicidal. It is effective within 30–40 minutes.

Control of Infectious Diseases

The infection control involves protection of patients and members of dental team from different kinds of infectious diseases. The procedures for infection control are designed to kill/remove microbes and to protect against contamination. One must follow 'infection control assurance', i.e. taking all steps necessary to ensure the desired results.

The goals for infection control program are:
- Decrease the number of pathogenic microbes to the level where normal body immune mechanisms can prevent infection.
- Eliminate cross-contamination.
- Treat all patients and instruments as though they could transmit an infectious disease.
- Protect patients and personnel from infection.

Patient Evaluation

The identification of patients with transmissible diseases and of those belonging to high-risk groups is essential before evaluation begins. A thorough medical history should be recorded and evaluated. The medical history serves the following purposes:
- Detect any recognized illness that requires medical diagnosis and treatment.
- Identify any infection or risk that may be important to a person exposed during examination/treatment.
- Manage the infected patients.
- Follow adequate infection control procedures, bearing in mind that general history-taking is not capable of detecting all infectious persons.

The medical history should be updated at subsequent visits. Specific questions should be asked regarding medications, current and recurrent illness, unintentional weight loss, lymphadenopathy, oral soft tissue lesions, other infections and history of hepatitis or any allergic reactions. Since the medical history and examination cannot identify all patients with blood-borne pathogens, the necessary precautions should be consistently used for all patients.

METHODS FOR INFECTION CONTROL

A. Immunization

Vaccines play an important role in the infection control process; hepatitis B is a major health hazard for dental health care personnel. Because of this risk, ADA Council on Dental Therapeutics and Centre for Disease Control (CDC) have recommended that all dental personnel should receive hepatitis B vaccine.

There is no vaccine for blood-borne pathogens, viz. HIV and non-A/non-B hepatitis. Proper infection control procedures are recommended to prevent transmission of such pathogens.

B. Personal Barriers

OSHA (Occupational Safety and Health Agency) and CDC (Centre for Disease Control) have recommended use of personal barrier as follows.

a. *Hand Washing*

The hand washing procedure begins with scrubbing all surfaces of the nails, fingers, hands and lower arms with an antimicrobial preparation. All jewellery must be removed. The initial scrubbing should be accomplished with a soft, sterile brush or a disposable sponge, each followed by a one or two minute rinse with cool/luke warm water.

The water should flow from fingertips to the elbow and not run backwards. Using a separate paper towel for each hand, drying should begin at the fingers moving to the hands and then the arms.

Hands must be washed before gloving. Washing hands before gloving reduces the skin microbial flora and helps prevent the irritation by waste products of bacterial growth under the gloves.

Hand cleansers containing a mild antiseptic like 3.0% PCMX (P-chloro, meta-xylenol) or chlorhexidine are preferred to control consistent pathogens and to suppress over growth of skin bacteria.

An alternative to hand scrubbing is a device employing pulsating water and soap lavage in a chamber where the hands are cleaned in 90 seconds.

b. *Gloves*

Gloves provide protection from contamination of micro-organisms on the operator's hands. Gloves need to be changed between patients and are not to be washed with detergents at any time.

It is recommended that gloves should not be reused. Double gloves may be indicated for patients with known infectious diseases, such as herpes virus,

Transmissible diseases of concern to dental surgeons and auxiliaries

- Hepatitis (type A, B, non-A/non-B)
- Acquired immune deficiency syndrome (human immunodeficiency virus): AIDS
- Syphilis
- Gonorrhea
- Influenzas
- Acute pharyngitis
- Pneumonias
- Tuberculosis
- Herpes
- Chickenpox
- Infectious mononucleosis
- Rubella
- Mumps

Individuals at high risk of contacting hepatitis B

- Health care personnel
- Selected patients and patient contacts
  - Patients and staff in hemodialysis units and hematology/oncology units
  - Patients requiring frequent or large-volume blood transfusions or clotting factors (i.e. hemophiliac patients)
  - Residents and staff of institutions for the mentally handicapped
  - Household and sexual contacts of persons with persistent hepatitis B antigen
  - Newborns of hepatitis B surface antigen carrier mothers

Populations with high incidence of hepatitis B

- Alaskan natives
- Indo-Chinese refugees
- Haitian refugees
- Native Pacific Islanders
- Sub-Saharan Africans
- Morticians and Embalmers
- Blood bank and plasma bank workers
- Prisoners
- Frequent user of illicit injectable drugs
- Frequent travellers

HBV and HIV. Gloves that have been contaminated with an infectious entity should be sterilized before being discarded. Boxes of gloves should be stored away from sunlight in tight, heavy plastic bags to minimize oxidation.

Criteria for selection of gloves include:

- No manufacturing defect
- Well-fitting
- Good tactile sensitivity
- Do not cause hypersensitivity
- Do not become tacky after wetting
- Resistant to wearing
- Conducive to glove powder
- Taste and odor not offensive
- Reasonable cost

Types of gloves

i. *Sterile surgical gloves:* These gloves assure a fit of high quality; however, they are expensive.

ii. *Latex gloves:* These are adversely affected by alcohols, sterilizing/disinfecting solutions, soaps and detergents. Washing latex gloves with soaps or detergents brings lipids to the surface resulting in their becoming tacky or sticky. Brown milled rubber gloves (thinner than latex and more fragile) are available for individuals who are allergic to latex.

iii. *Vinyl gloves:* Referred to as 'overgloves', these are used when intraoral procedure has to be interrupted for a brief time, e.g. to answer a telephone call or briefly examine another patient. The overgloves can be slipped over the regular gloves and removed when work with the initial patient is resumed.

iv. *Heavy utility gloves:* These should be worn when handling contaminated sharp instruments because they are wear resistant. Certain utility gloves (nitrite types) can be washed, sterilized and are tear resistant.

Gloving procedure: Gloving may be started by pulling the glove into left hand first. Using the right hand, the cuff (inner surface) of the left glove is grasped and the glove pulled on. The right hand glove is then picked up by inserting the fingers of the left glove inside (outer surface) of the rolled cuff of the right glove. The right hand is inserted into the glove and the glove pulled into place. Modified starch powder has replaced talcum as the dusting agent of choice. Sterile creams can also be used.

c. Face Masks

The face mask is an important barrier providing protection from inhalation of aerosols, generated by high-speed handpiece and air water syringes. It also prevents spatter from patients' oral cavity and splashes of contaminated solutions from contacting the mucous membrane of oral cavity and nose. Face masks may be composed of glass or synthetic fiber, paper, cloth, foam or other synthetic materials. The fiber type mask is considered to be more efficient in filtering bacteria. Effective face masks should provide a minimum filtration of 95% and should have ability to block aerosols as well as larger particles of blood, saliva and oral debris.

A properly fitting mask should:

- Fit comfortably around the entire periphery of face
- Not leak air out of the sides
- Not touch lips or nostrils
- Not irritate skin
- Provide breathing effectively
- Not cause fogging of protective eyewear

Masks with highest filtration are rectangular and folded types, mainly used for surgeries. Dome-shaped masks are adequate barriers against spatter and are considered to prevent HBV and HIV infection. To protect against aerosols, edges of the rectangular mask are pressed around the bridge of nose and face. Masks should be grasped only by the string or band at the sides or back of head to remove it.

Face masks should be changed once per hour or between each patient contact or whenever it becomes moist or visibly soiled.

The outside of masks should not be touched as it significantly decreases the filtration capability of mask.

d. *Protective Clothing*

The protective clothing includes reusable/disposable gowns, laboratory coats with long sleeves, high neck and long knee length. Head covers are also recommended during procedures that result in splashing blood or other body fluids as hair can trap heavy contamination and if not washed thoroughly, can be rubbed back from a pillow onto the face at night.

The gowns should be changed at least daily and should be changed immediately, if soaked or spattered with blood or other contaminants. The reusable garments should be placed in bags marked with a 'biohazard' symbol. Laundering can be effectively accomplished with a high temperature (60°C to 70°C) wash cycle with normal bleach followed by machine drying (100°C or more). Cool water containing 50 to 150 ppm of chlorine can also be used. This method has been found to be effective in killing the AIDS virus.

The slippers in operatory should also be cleaned/disinfected, at least once a day.

e. *Eyewear*

Protective eyewear is highly beneficial for dental personnel and for the patient. Herpes virus infection of eye and infection from hepatitis B are possible consequences of viral contact with eye.

Put an eyewear with clean hands before gloving and remove it with clean hands after gloves are removed. When eyewears are removed, they should be cleaned and disinfected. After removal, eyewear should be placed on a paper towel and sprayed with a water-based disinfectant. Wash eyewear well, reapply disinfectant for 10 minutes, or preferably allow it to soak in a 1:50 to 1:100 solution of 5.0% hypochlorite or other disinfectant solution that does not affect eyewear. Rinse well and dry.

Plastic lenses, if used, should be immersed in 20% alkaline glutaraldehyde solution and then thoroughly rinsed.

C. Surgical Asepsis

a. *Surgical Staff Preparation*

The staff in a surgery (operatory) should be prepared as follows:

- Street clothes are replaced with a scrub consisting of clean linen trousers and short-sleeved blouse. Conducive footwear should be used.
- The operators and paramedical staff moving in an operating room should cover their hair.
- Face mask must be worn. Sneezing and coughing are not permitted in the operation room.
- The nails should be checked for cleanliness. All gross sub-nail contaminations should be removed with a nail cleaner.
- The arms are then wetted to several inches above the elbows. The hands are kept elevated above elbow height so that the direction of water flow will be towards the elbows.
- The surgeon is helped into the sterile gown by a properly gowned and gloved assistant. A circulating assistant secures the gown ties at surgeon's back. The surgeon's back as well as area below waist are considerably unsterile.

b. *Preparation of the Surgical Site*

Preparation of surgical site is not of importance in conservative dentistry; only for procedures employed on skin.

The operative site may not be totally sterilized, but gross cleaning action reduces the incidence of post-operative infections.

The iodophor compounds are the most effective solutions for the skin preparation.

If there are hair in the surgical area, it is removed just prior to scrubbing the skin. (Shaving the area the night before may produce small abrasions on the skin and resident bacteria can multiply as a result of the injury.)

The scrub should begin in the center of the area and then move outward concentrically. Once the center has been scrubbed, it should not be touched again with the same sponge.

c. Draping the Patient

A double layer drape is considered adequate. The patient's head is wrapped with double sheet, using a drape as the lower sheet and a hand towel as the upper sheet.

Sterile drapings are secured with towel clips. The anesthetist and his or her equipment are isolated from the operating team by drape-covered screen.

A gown, drape or covering is considered to be contaminated when wet, unless they are waterproof or otherwise backed by a waterproof lining.

D. Protection against Aerosols and Spatter

The rubber dam has been shown to be an effective barrier to reduce the number of organisms in aerosols. The rubber dam prevents aerosolization of saliva and should be used whenever possible.

In addition to rubber dam, high volume evacuator and low volume saliva ejector help in minimizing contact with aerosols and spatter. While using low suction saliva ejection, caution is needed to reduce chances of back flow into patients' mouth.

Patients rinsing their oral cavities with an antimicrobial solutions just before the treatment has been shown to reduce the number of bacteria that exit the oral cavity in the form of aerosols and spatter during operative procedures.

E. Surface Asepsis

Micro-organisms can survive for prolonged period on instrument surfaces, particularly when associated with saliva or blood.

The dental equipment must be disinfected/sterilized in between patient visits.

Surface asepsis includes:

- Clean and disinfect contaminated surfaces
- Prevent the surface from being contaminated (use of surface barriers).

The general principle is to use barriers, which are manufactured from a material that is impermeable to water, blood, saliva and microbes.

The barrier can be of laminated paper-plastics, treated paper/fabric or aluminum foil.

a. Instrument Delivery

Instrument delivery implies shifting of sterilized instrument to the site of operatory. The surfaces should be cleaned and disinfected before the next patient is attended. A disinfectant should be applied to the contaminated surface either by spraying or by using a saturated pad.

The effective method of instrument delivery is through the use of barrier materials. Non-permeable single layer covers and barriers should be used during each patient.

- Plastic food wraps, plastic bags, aluminum foil or custom made barriers provide an inexpensive method of protection from local contaminants.
- Special attention must be paid to equipment knobs, switches and handles that are touched during treatment. These items should be covered individually with wraps.
- Handpiece hangers are subjected to direct contamination and must be carefully disinfected between patients.
- The edge of unit heads is especially prone to contamination when grasped with fingers during movement. Equipment should be moved only with handles, which are easily disinfected and/or protected with barrier materials.

b. Dental Chair

The dental chair including upholstery, should withstand repeated disinfection. The upholstery should be removable for cleaning and replacements. The frame should be of non-organic material, that does not support the growth of pathogens. All chair functions should be controlled from a foot switch to avoid possible contamination by use of hand operated switches.

The potential for cross-contamination in a dental chair is from chair mounted controls. The finger-operated switches normally have cracks and crevices, which are difficult to clean and disinfect.

The headrests should be covered by disposable covers. Arm slings tend to trap debris and must be carefully cleaned prior to disinfection.

c. Operator Stool

Adjustment of the height of the stool is usually accompanied by grasping a lever located under the stand. Covering the lever with plastic sheath will control cross-contamination.

The seat material should be of vinyl plastic, particularly when maximum asepsis is required. Cleaning and disinfecting the seat covering may be accomplished with soap and water.

d. Cabinet

The materials and equipment should be stored out of the treatment room and brought in only when needed for a particular procedure.

The cabinets should be made of materials that will withstand repeated cleaning and disinfection. Work surfaces should be disinfected for each patient. Other surfaces can be disinfected twice a week.

e. *Utility Items*

A potential source of physical and microbial contamination is the main water supply. Preventive measures for water systems include use of a water sediment filter and softening and/or deionization of the incoming water supply, which will help prevent clogging of water passages in dental equipment.

The sediment trap on incoming line is a real source of contamination. The trap should be located in an accessible area. The trap should be disassembled and flushed with copious amounts of water to remove all sediments and debris collected in the cup.

The air compressor intake filter must be positioned in an area of clean, cool and dry air. Care must be taken so that the intake filter is cleaned and replaced on a regular schedule.

Air compressor tank must be drained periodically to ensure that water is not building up in the tank. This water is a source of microbial contamination.

f. *Dental Unit Water Lines*

It has been established that the dental unit water lines are usually contaminated.

Contaminated oral fluids may be drawn back into turbine chamber of handpiece or may be retracted into the water lines. While dental units have anti-retraction valves, these valves do fail. The retraction should be checked periodically by observing the tip of waterline proper. If a drop of water 'hangs' on tip, retraction is not adequate.

The dental unit waterlines become colonized with micro-organisms including bacteria, fungi and protozoa, which form the biofilm. The micro-organisms in the biofilm produce a protective polysaccharide matrix that serves as a reservoir, which can amplify the number of micro-organisms. These micro-organisms exit through dental handpiece or air water syringe.

The ADA recommends that infection control measures be followed, such that dental unit should contain <200 CFU/mL (colony-forming unit/mL).

The dental waterlines should be flushed at the beginning of the day to decrease the microbial load. However, this does not affect biofilm in the waterline or improve the quality of water used during dental treatment. The effective methods include self-contained water systems combined with chemical treatment, and in-line microfilters.

The blood and saliva with micro-organisms can enter the dental water system during patient treatment.

The majority of dental units are engineered to prevent retraction of oral fluids. Even with anti-retraction valves, flushing for a minimum of 20–30 seconds after each patient is recommended.

Sterile solutions are used as a coolant/irrigant performing surgical procedures. Conventional dental units cannot deliver sterile water even when equipped with independent water reservoirs because the water bearing pathway cannot be reliably sterilized. Delivery devices, preferably disposable products, should be used to deliver sterile water.

g. *Saliva Ejectors*

Back flow from low-saliva ejectors occurs when the pressure in the patient's oral cavity is less than that in the evacuator. The micro-organisms present in the line usually get retracted into the patients' oral cavity. This back flow can be a potential source of cross-contamination.

The suction tubings should be smooth and even on both internal and external surfaces. A few systems are available with removable tubings, which may be heat sterilized; however, most systems need disinfection procedures.

Tubings for handpieces, scalers, air/water syringes and other instruments are difficult to maintain in an aseptic condition.

Cameras, light-curing units, lasers, air abrasion units, etc. are miscellaneous items that must be protected against contamination. They are used in operatory and cannot be sterilized or even readily disinfected. Clear plastic bags of suitable size should be used to cover these items.

The researchers are working hard to improve the sterilization and asepsis protocol of the dental equipment. Molded, seamless components are being designed and manufactured for all types of treatment equipment. Non-contact controls are being popularized for achieving better asepsis.

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